NPTEL - Theory of Computation

Assignment - Week 2

Question 1 (2 points)

Which of the following is equivalent to the set described by the regular expression 11^*00^*?

A. $\{1^m0^m | m > 0$ is a natural number$\}$

B. $\{1^m0^n | m, n > 0$ **are natural numbers**$\}$

C. $\{1^m0^m | m \geq 0$ is a natural number$\}$

D. $\{1^m0^n | m, n \geq 0$ are natural numbers$\}$

Option B is correct. There can be any number of 1's and 0's in the strings belonging to this language. It is not necessary that the number of 0's and 1's be equal. But there should be at least one 0 and one 1 in the string.

Question 2 (2 points)

Let L be a regular language. Then what can we say about the language $L' = \{a_n a_{n-1} \ldots a_1 \mid n \geq 0, a_1 a_2 \ldots a_n \in L\}$?

A. L' **is regular**

B. L' is not regular

C. L' may or may not be regular

D. L' cannot be generated by a regular expression

Option A is correct. Refer to Closure Properties of Regular languages for Reversal (Section 5.1.2, Week 2 Lecture Notes)

Question 3 (2 points)

Any regular expression of size n can be converted to an NFA (with ϵ-transitions) having at most (select the smallest possible asymptotic upper bound)

A. $O(n)$ **states**

B. $O(n^2)$ states

C. $O(2^n)$ states

D. $O(1)$ states

Option A is correct. Refer to Converting an RE to NFA (Section 4.2.1, Week 2 Lecture Notes). For each symbol in the regular expression, we need to add at most 2 states to the corresponding NFA (given that we allow ϵ transitions). We also need to add at most 2 states to the corresponding NFA for each concatenation. There can be at most $n - 1$ concatenations in a regular expression with n symbols. Therefore, an NFA can be constructed corresponding to a regular expression of length (size) n with at most $4n - 2$ states, which is $O(n)$.

Question 4 (2 points)

Which of the following regular expressions generates the following language?

$$L = \{w \in \{0, 1\}^* \mid w \text{ begins with 1 and ends with 0}\}$$

Option A is correct. $(0 + 1)^*$ is the regular expression for the universal language. Therefore, $1(0 + 1)^*0$ is the regular expression for the language of strings beginning with 1 and ending with 0 (and having anything in between).

A. $1(0 + 1)^*0$

B. $(10)^*$

C. 1^*0^*

D. $1(01)^*0$

Question 5 (2 points)

Which of the following regular expressions is equivalent to the regular expression $(1+\epsilon)(01)^*(0+\epsilon)$?

A. $(0+\epsilon)(01)^*(1+\epsilon)$

B. $(1+\epsilon)(10)^*(0+\epsilon)$

C. $(0+\epsilon)(10)^*(1+\epsilon)$

D. $(01)^* + (10)^*$

Option C is correct. The key observations we make here are $1(01)^* = (10)^*1$ and $(01)^*0 = 0(10)^*$. Also we note that $(01)^* = 0(10)^*1 + \epsilon$ and $(10)^*10 + \epsilon = (10)^*$. We now use the commutativity of $+$ and distributivity of concatenation over $+$ in regular expressions and get the following derivation.

$(1+\epsilon)(01)^*(0+\epsilon)$
$= 1(01)^*0 + 1(01)^* + (01)^*0 + (01)^*$
$= (10)^*10 + (10)^*1 + 0(10)^* + 0(10)^*1 + \epsilon$
$= ((10)^*10 + \epsilon) + (10)^*1 + 0(10)^* + 0(10)^*1$
$= (10)^* + (10)^*1 + 0(10)^* + 0(10)^*1$
$= (10)^*(1+\epsilon) + 0(10)^*(1+\epsilon)$
$= ((10)^* + 0(10)^*)(1+\epsilon)$
$= (0+\epsilon)(10)^*(1+\epsilon)$

Question 6 (2 points)

Which of the following regular expressions generates the complement of the language generated by the regular expression $(00+11)^*$?

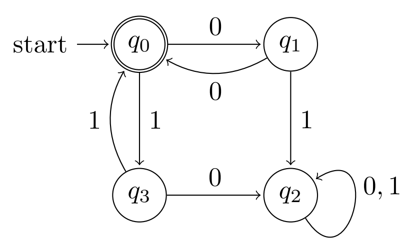
A. $(00+11)^*(0+1) + (00+11)^*(01+10)(0+1)^*$

B. $(00+11)(0+1)^* + 0(00)^* + 1(11)^*$

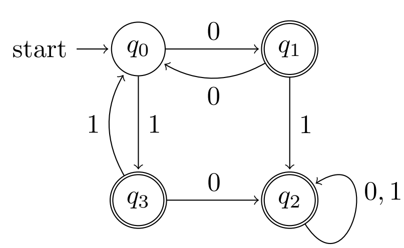
C. $(01+10)(0+1)^* + 0 + 1$

D. $(00+11)(0+1)^* + (0+1)(01+10)^*$

Option A is correct. First construct the DFA for the given regular expression.

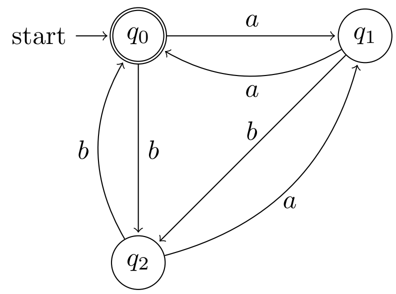


Then, construct the DFA for the complement language by making the accepting states in the DFA as non-accepting and the non-accepting states as accepting.



Now convert this DFA to regular expression following the process as described in Section 4.2.3 of Week 2 Lecture Notes.

Question 7 (2 points)
Which of the following regular expressions generates the same language as the given DFA?



A. $(a(b+\epsilon)(ba)^*a + b(a+\epsilon)(ab)^*b)^*$

B. $((b+\epsilon)a(ab)^*a + (a+\epsilon)b(ba)^*b)^*$

C. $((b+\epsilon)a(ba)^*a + (a+\epsilon)b(ab)^*b)^*$

D. $(a(a+\epsilon)(ba)^*a + b(b+\epsilon)(ab)^*b)^*$

Option C is correct. Convert the DFA to regular expression following the process as described in Section 4.2.3 of Week 2 Lecture Notes.

Question 8 (2 points)
Let R and S be two regular expressions. Which of the following is equivalent to the regular expression $S(S+RS)^*$?

A. $R(SR+R)^*$

B. $(RR^*S^*S^*)^*$

C. $(S+SR)^*R$

D. $RR^*S(RR^*S)^*$

None of the options is correct. This question will be reevaluated. That is, everyone will get marks for this question.

There is a typo in Option C, which was supposed to be the correct answer. Option C was intended to be $(S+SR)^*S$.

For any two regular expressions P and Q, $(PQ)^*$ can be written as $P(QP)^*Q+\epsilon$. Using this principle and the commutativity of $+$ and distributivity of concatenation over $+$ in regular expressions, we have the following derivation.

$S(S+RS)^*$
$= S((R+\epsilon)S)^*$
$= S((R+\epsilon)(S(R+\epsilon))^*S+\epsilon)$
$= S(R+\epsilon)(SR+S)^*S+S$
$= (SR+S)(SR+S)^*S+S$
$= ((SR+S)(SR+S)^*+\epsilon)S$
$= (SR+S)^*S$